

Kevin Rivera

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EDUCATION

Duke University

Master of Engineering in Robotics & Autonomy

August 2026 – Present

University of Central Florida

Bachelor of Science, Mechanical Engineering | GPA: 3.51

May 2022 – May 2026

EXPERIENCE

Agricultural Robotics Lab – UCF

Volunteer Undergraduate Research Assistant

May 2025 – August 2026

- Developed a novel, low-cost depth estimation system for a strawberry-harvesting robot using a decoupled dual RGB camera setup and SSE-minimized polynomial model; achieved 0.443 mm RMS accuracy (~1% normalized error) without RGB-D cameras, LiDAR, or neural networks
- Designed the full real-time MATLAB image processing pipeline: weighted RGB decomposition, binary thresholding, morphological operations, kernel smoothing, and fruit perimeter extraction
- Modeled SolidWorks test fixtures and end-effector mounts with modular architecture to minimize reprinting costs; collaborated on Raspberry Pi integration for dual-camera and motor control
- Co-authoring a research paper for journal/conference publication; delivered weekly progress reports as one of two founding members of the research effort

Jet Set Investments LLC

Product Development Engineer

March 2025 – June 2026

- Developed and contributed to a company patent for innovative roadway lighting solutions decreasing cost while improving safety and energy efficiency.

Hubbard Construction (Vinci)

Estimating Intern

May 2024 – January 2025

- Produced cost analyses and estimates for asphalt projects serving FDOT, Walt Disney World, Universal Studios, Orlando International Airport, and multiple Florida counties

Dr. Sohn Materials Science and Engineering Research Group (UCF)

Undergraduate Research Volunteer

November 2022 – May 2024

- Prepared and tested 3D-printed samples for optimal printing parameters; contributed training data for an AI material-property prediction tool
- Operated advanced equipment: SEM, SLM printer, and gas atomization systems

PROJECTS

UCF Senior Design Project - Bring ROAR to Life

Anatomically correct skeleton: animatronic Stegosaurus, from concept to fully implemented prototype

August 2025 – Present

- Developing an animatronic Stegosaurus; designed electrical systems, wiring diagrams, and teleoperated control algorithm; full cost estimate analysis; co-authored multiple 150+ page design reports with stakeholder and risk analysis
- Collaborated with the London Natural History Museum (Matthew Dempsey) and Lawrence Hall of Science (UC Berkeley) on skeletal accuracy, theoretical customer design feedback, and a potential educational donation; weekly design reviews with an adjunct NASA professor

Wizardry Chess

Built a fully automated chess board synced to live Lichess games via API; custom XY gantry, stepper motors, and electromagnet actuation controlled by a Raspberry Pi with ROS2

January 2026 – May 2026

- Implemented a "Loyalty Mechanic": live engine analysis triggers insubordinate random piece moves when a player blunders, adding an experimental adaptive game layer

LEADERSHIP

Canciller & Former Secretary Delta Chapter Non-Profit - Phi Eta Mu

Led operations and administration of a 40-member chapter; organized cross-chapter events including Universal Halloween Horror Nights and a year-end camping trip (80+ attendees), increased chapter engagement by 20%, created and sold merchandise that increased chapter funds by 60%

August 2024 – Present

- Founded the Phi Eta Mu Networking LinkedIn group with supreme council support to strengthen alumni engagement and career development across chapters

SKILLS

Software: MATLAB, Python, C++, AutoCAD, SolidWorks, Fusion360, HCSS HeavyBid, BlueBeam, Photoshop, Canva, Inkscape, Microsoft Suite, ImageJ, Fiji

Lab Instrumentation: Laser melting printers, gas atomization, metal polishing, SEM, XRD, Scientific Furnace, Arduino, Raspberry Pi (Pico), ESP32

Languages: Spanish and English